

A Decision Making Approach for Street Lockdown to Cope with Covid-19 Cases by Using Shortest Path Selection Mechanism for Unplanned Colonies

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Abstract— Infectious disease syndrome like covid-19 falls under the Public health domain and needs to be addressed with timely decisions and rapid actions. For such diseases, the dispersal becomes exponential with frequent social gatherings, therefore the immediate strategy, to control the surging waves of covid-19, was to impose immediate lockdown of COVID-19 infected zones. In this paper, the concept of street networks has been incorporated with shortest path algorithm e.g. minimum spanning tree (MST) to define an approach to investigate the correlation between reported COVID-19 cases and relevant streets in order to adopt better lockdown strategy for unplanned colonies. Geo-spatial representation has been used for subsequent composition of patterns to identify the particular streets for locked down. Results show that MST provides better solution by evaluating explicit areas of concern for lockdown plans.

Keywords—Spatial Networks, COVID-19, Public health, Infectious Diseases, Street Networks, Unplanned Colonies.

I. INTRODUCTION

The Covid-19 pandemic came with numerous challenges to social as well as economic practices resulting in so many barriers to national and international development. Being the most devastating calamity in the history, COVID-19 has intimated many key areas of concern e.g. role of design and planning in urban systems. These factors can be considered as pivots to deal with the epidemics. As immediate reaction of outbreak at hand, the very first and effective strategy adopted in worldwide nations was the lock down of workspaces, public places and residential areas. For residential areas, the lock down strategy has been adopted for immediate closure of streets coming out with reported cases of COVID-19. To deal with such alarming scenarios, the concept of street networks comes with the insight of better planning and actions to be taken.

The concept of street network has been used widely to examine its role in finding different socio-economic patterns. It comprises of the paths adopted to move between two specific locations which in turn can be used to find out the mobility behaviour to define the patterns for movement. A Segment of street, defined as a section connecting two junctions, comprises the basic unit of a street network. There are two ways for representation of street networks. In dual representation, roads are considered as nodes and on the basis of sharing of junction,

link is placed between the two nodes. For primal representation, the concept has been formulated by placing the link between two nodes connected by the road, where nodes represent the junctions. To model the physical infrastructure of streets, graphs have been used to present the abstract modelling which is useful in various analysis and investigations. This breaks down the complex representation of street network by defining the objects as nodes, also called vertices, and links as edges. This concept can be defined as network modelling which is a subject of interest while exploring street networks. In order to solve daily life problems the popularity of techniques, for finding the shortest path, has been amplified. As the result of shortest path, diverse aspects have been found to provide services that have been proved fruitful. In context of finding shortest path, minimum spanning tree (MST) has been used extensively for diversity of networks as an optimization problem, so that a tree structure can be found by exploring the minimum number of edges connecting the nodes provided with minimized cost.

Concepts of spatial networks and minimum spanning tree have been used widely to investigate various issues in literature, but bringing them together in combination to assist the lockdown strategy is a quiet new dimension. This research paper targets the utilization of spatial networks in order to devise an approach to explore the street patterns, in context of COVID-19 cases, with application of the shortest path mechanism. Purpose is to find the targeted areas that can be locked down so that spread can be controlled in timely manner.

Rest of the paper has been accumulated as follows. Section II gives the overview of related work in perspective of three domains: COVID-19, Spatial Networks and Minimum Spanning Tree. Methodology has been covered in Section III. Section IV comes with the result and analysis. Section IV carries the conclusion and future work.

II. BACKGROUND

Over the past year, many research perspectives have been explored with the outburst of COVID-19. Proficiency of urban systems to deal with pandemic, in perspective of socio-economic factors e.g. economy, infrastructure and environment etc., has become an intensive research trend. In order to cope with the pandemic, urban resilience is considered in perspective

of socio- ecological models to improve the environment that can help to minimise the effects of a pandemic [1]. To achieve this, identification of ecological, social and economic factors for a particular atmosphere/environment is needed. Considering the factors of an atmosphere in narrow and wide range of exposure can influence the blowout of a disease and must be addressed in exertions to combat the epidemic [2]. By keeping in view the great share of time being spent at home, surroundings and working environment, effects on public health during the giant wave of infectious disaster can be better explored. Timely control and reduction of these effects can be determined with effective policymaking strategies. The number of indicators, sustaining the pliability of a pandemic, and their relationship with the surrounding areas have been investigated in order to provide an index to indicate the potential to deal with the pandemic in various dimensions e.g. socio-economic, infrastructure and environment [3]. For Yangtze River Delta, different factors like economy, ecology and infrastructure, in perspective of achieving the potential to cope with pandemic like COVID-19, have been explored. Different frameworks and indicators can be used in order to achieve the potential to deal with disasters. In order to improve the resilience of urban system an index system to evaluate the resilience has been proposed with consideration of the economical, ecological and infrastructural aspects of a selected area [4].

Infrastructure of residential areas can be explored using the concept of street networks in order to investigate underlying social issues. Currently phenomenon of street networks has been used widely to figure out the crime ratio by utilizing the residential addresses and investigating the conventional statistics of streets. Betweenness value for streets has provided granularity to find out the streets that were more disposed to crime [5]. To model the physical infrastructure of streets, graphs are used to present the abstract modelling which is useful in various analysis and investigations. Concept of spatial network can be defined as a representation of transport network where linear elements are connected in a given space. Movement on a

path from one point to other along with associated cost is the point of concern [6].

There is a possibility that, for a given source and destination, all shortest paths explored by k shortest path problem might be sharing a wide numeral of edges [7], [8], [9]. Physical infrastructure has been used to explore the spatial distribution in context to examine the violence in terms of murder etc. Streets with high value of betweenness need to be focused to take measures to reduce the crime [10]. Methods based on street networks can outperform for future prediction of unusual activity e.g. crime [11]. Exploring the shortest paths for given street network to find out the street segments of interest, has become an object of great interest in mining the different patterns and trends. MST has the potential to dig out the connectivity between the residential patches [14]. By utilizing concept of the street networks, a protection strategy for arterial passageways has been proposed. Major passageways from street network have been identified by using MST with consideration of the expansion in urban areas and natural boundary in order to evaluate the stability of ecological network [15]. Concept of spatial networks has been utilized with spanning tree by means of Euclidean distance to discover social practices in order to evaluate different groups [16]. An algorithm based on MST has been proposed to connect the sets of events in such a way that sum of edges, connecting the events in a tree, has the minimum value [17]. Dijkstra algorithm yields the shortest paths in iterative manner. Previously, it has been used frequently in GIS applications and graph theory. It takes the source as centre point to search outwards iteratively while expanding to get the target also called destination [18].

III. CASE STUDY

Multiple units, executing parallel at ministry of health deal with outbreaks in different regions. Unit manager provides the services to deal with reported problem after collecting information and analysing the problem. In order to deal with the

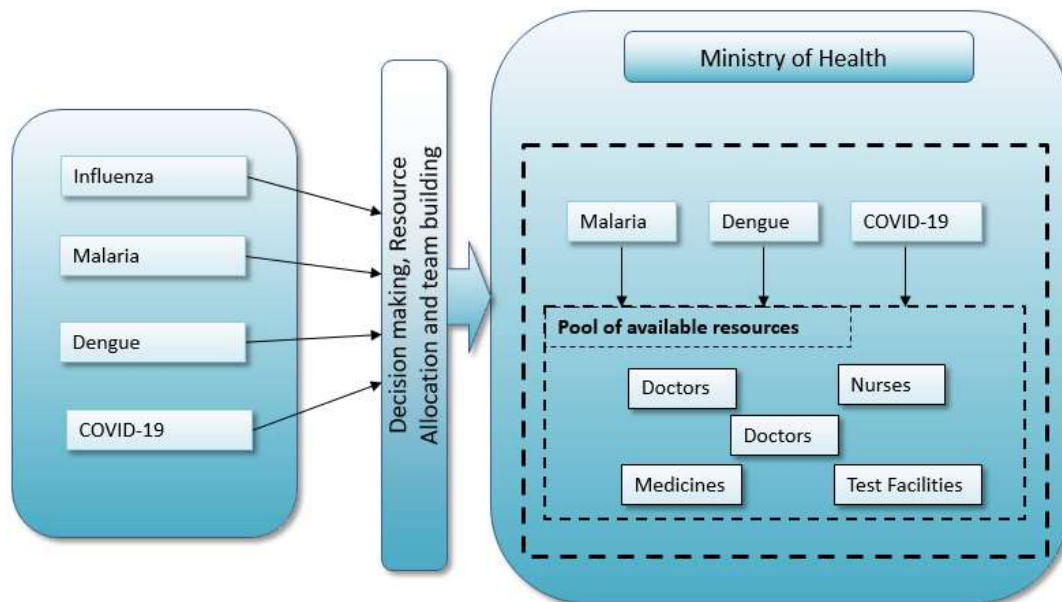


Fig. 1. Public health programs dealing with different outbreak

epidemic, the concerned manager has to take decisions about resource allocation in terms of medical equipment, staff and medicines. Figure 1 describes the functionality of multiple units. Concerned authorities can adopt strategic policies as well in parallel to resource allocation.

IV. METHODOLOGY

The proposed approach employs the concepts of street network for unplanned colonies to implement an effective decision making policy for lockdown of streets based on reported number of cases. A specific sector in capital territory of Pakistan has been captured for the implementation of proposed model. Infrastructure of the capital city comprises of different sectors that are well organized and have planned colonies that come up with proper addresses containing streets and house numbers respectively. Apart from this some sectors have completely unplanned infrastructure. There exists no such organization of house addresses for unplanned colonies. One of these has been picked for execution of proposed approach.

Figure 2 represents the steps for implementation of the proposed approach. Data comprises of the house address instances pointing the victims of COVID-19. These addresses need to be refined in order to get complete information, therefore, after data processing geographical representation was used to provide a clear insight into areas that have been marked for reported cases. House addresses were used to obtain geographical representation of unplanned areas. Next step involved finding shortest path of connected nodes in order to highlight pandemic-hit areas of concern. . Whenever a victim is

identified in certain area, an exponential blowout of disease is considered imminent in that particular area. At this point the algorithms for shortest path discovery are considered optimal for investigation of spread of Covid-19 in pandemic inflicted area. Target areas can be explored as a result of exploration of shortest path. To assist the lockdown policy, identification of strategic information played a vital role in order to take the effective measures. Furthermore, to find a certain pattern, MST was opted to find the tree structure connecting the streets, highlighting affected houses that could be locked down immediately. List of areas was achieved in this manner and added to knowledge base for further processing. Clear vision through map is represented and hence lockdown strategy could be applied on particular streets for highlighted area. Table I presents the sample values used to mark the house addresses on map. For houses occupying the infrastructure such that it covers two streets, any street can be picked. Geospatial representation was obtained using concise longitude/latitude information.

TABLE I. LONGITUDE LATITUDE FOR HOUSE ADDRESSES

Node Id	Longitude/Latitude Values	
	House	Street
1	33.657682, 72.989011	72.988982, 33.657732
2	33.658007, 72.990343	72.990318, 33.658093
3	33.657666, 72.990471	72.990174, 33.657444
4	72.990142, 33.657524	33.657548, 72.990048

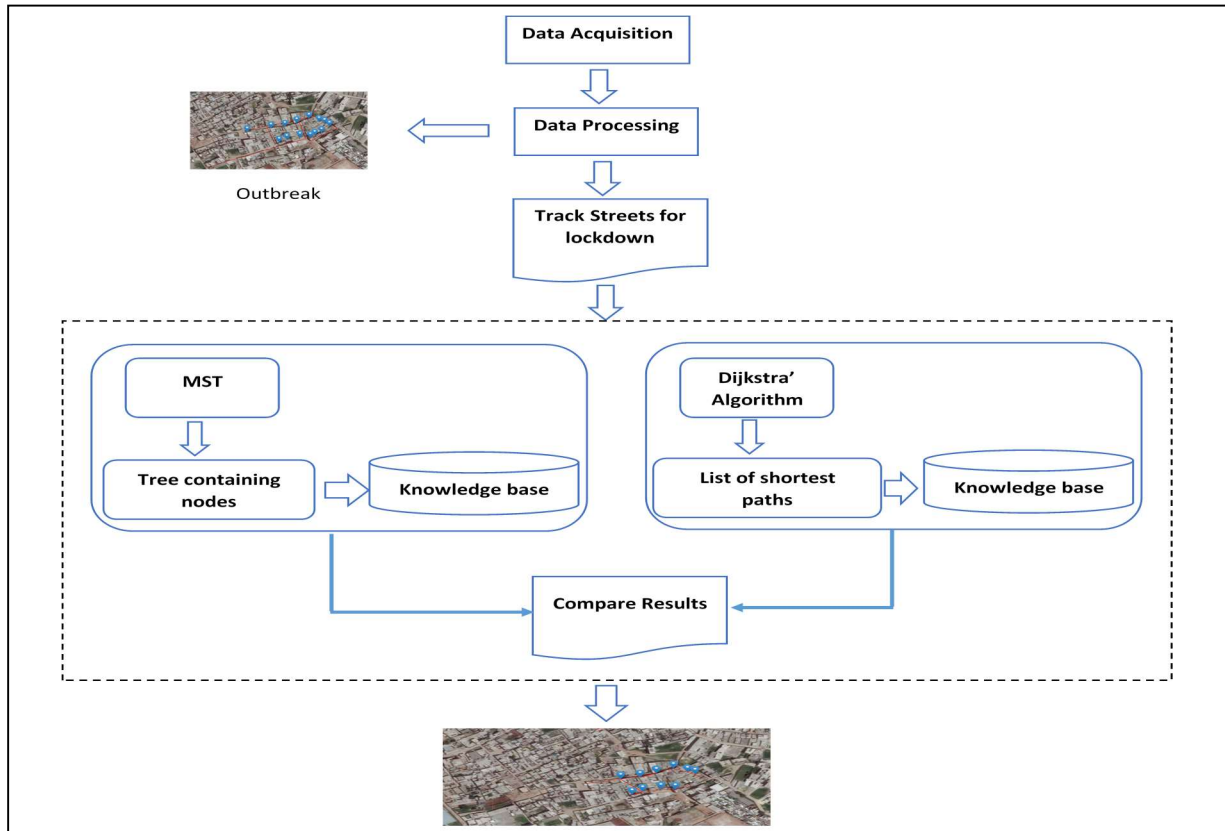


Fig. 2. Mechanism to find the targeted streets for lockdown

V. EXPERIMENTAL RESULTS

In order to get the optimal shortest paths, two algorithms have been implemented, namely, 1) Minimum Spanning Tree 2) Dijkstra Algorithm respectively. The algorithms for proposed approach have been implemented in Python. Firstly, the spatial representation of plausible COVID-19 victims has been obtained. This illustration has been presented in Figure 3, signifying the problem areas where streets and affected houses have been marked. Graph representation for the geospatial representation of the reported victims consists of nodes and edges where nodes were represented as houses connected by streets carrying the estimations of distances between them. Problem areas comprising of houses as nodes and streets as edges has been plotted using concepts of spatial networks. Geospatial data is important to trace the victims, hence utilized to get a clear insight into more vulnerable areas



Fig. 3. Selected points in terms of houses and streets

In Figure 4, steps for the proposed approach have been described which explains how the areas of concern were identified and highlighted on the map. First step towards the application of shortest path algorithms is to get an adjacency matrix for given graph, where weights represent the distances between the houses located on same street. Incoming data has been distinguished to find proper house addresses by applying the addressing scheme after that a proper geospatial representation was drawn. This assembly of houses and streets is presented as a graph. Adjacency matrix for given graph was used for further processing by application of shortest path algorithms.

Main input for application of MST is the adjacency matrix, which is achieved from the graph made up of victimized houses as nodes and relevant streets. Houses as nodes are presented in set H , whereas set S contains the streets as edges. A_{ij} represents the adjacency matrix that is main input for SPA module. Shortest path algorithms e.g. Dijkstra Algorithm and Minimum Spanning Tree Algorithm have been applied, while getting each node, a node set H_1 has been created that will hold all addresses appearing in the shortest path for given algorithm. Meanwhile geo-spatial information has been maintained with each node. $Val_{hlong/lat}$ represents a set containing geospatial information for each node. Getting this geospatial information is important for each candidate node sharing the shortest path. Module_{SPA} (module for shortest path algorithm) is representation of both minimum spanning tree algorithm and Dijkstra algorithm. Each

algorithm was applied stepwise to get list of shortest paths to examine the pattern of blowout.

Algorithm

INPUT: Street Network Graph Representation $G = (H, S)$

$H \rightarrow$ nodes representing houses

$S \rightarrow$ edges representing the streets

Steps

Convert $G = (H, S) \rightarrow A_{ij}$ (Adjacency Matrix)

$A_{ij} \leftarrow$ Module_{SPA} // Apply shortest path algorithm

While nodes found in SPA

$h_i \leftarrow$ construct H_1 // add h_i to (node set) $H_1 \leftarrow \{h_1, h_2, h_3, \dots, h_n\}$

$h_i \leftarrow Val_{hlong/lat}$ // Get values for geo-spatial position

OUTPUT in their representative geospatial position

Fig. 4. Algorithm to find the streets with COVID-19 victims

A. Implementation of Dijkstra's Algorithm

Dijkstra Algorithm needs a specific source node for its application in order to generate the shortest paths for all nodes in graph. Distance from the source point has been calculated from all the nodes in graph. Therefore, for a given graph, the results contain the shortest paths from the source node to all other nodes. Figure 5 shows the geospatial representation of results achieved by the application of Dijkstra Algorithm.

B. Implementation of Minimum Spanning Tree Algorithm

Minimum spanning tree digs the connected nodes in tree structure by providing the minimum sum of cost of all connecting edges. For a graph, it comes with the tree of connected nodes while omitting the loops. Results for minimum spanning tree algorithm have been represented in Figure 6. Streets containing the victims in form of tree of connected nodes have been visualized using spatial representation. A complete path was plotted to get the pattern of blowout. This imparted clear insight into the lockdown strategy of identified streets. Results show that a better lockdown strategy can be built upon the evaluations provided by the MST. An exact pattern of connection is required in form of a tree where nodes are connected.

Results have been provided to observe the working mechanism of Dijkstra algorithm as compared to MST in background of Covid-19 lockdown. Figure 7 shows that main focus of Dijkstra is to calculate distances and shortest paths for each node from source node. Source node can be selected randomly. The distance of various nodes from a given source node is not as fruitful in finding the single shortest path described for lock down, rather it provides the variety of shortest paths particular to every node in the graph. Whereas MST provides a single shortest path in form of spanning tree with associated costs. Figure 8 summarizes the results of MST where a tree comprises of all nodes that are connected, without having



Fig. 5. Spatial representation of minimum spanning tree



Fig. 6. Spatial representation of Dijkstra's Algorithm

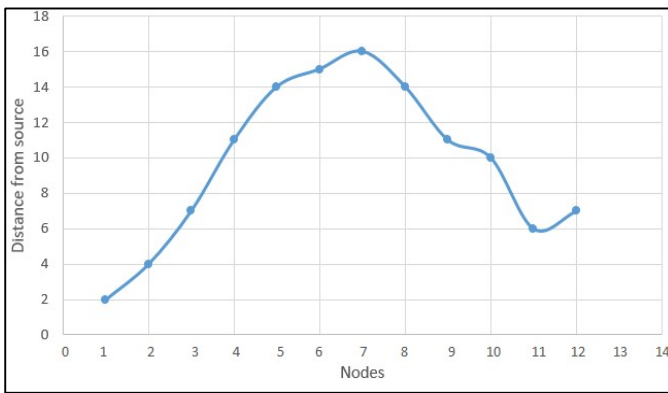


Fig. 7. Dijkstra Algorithm: Distance form source

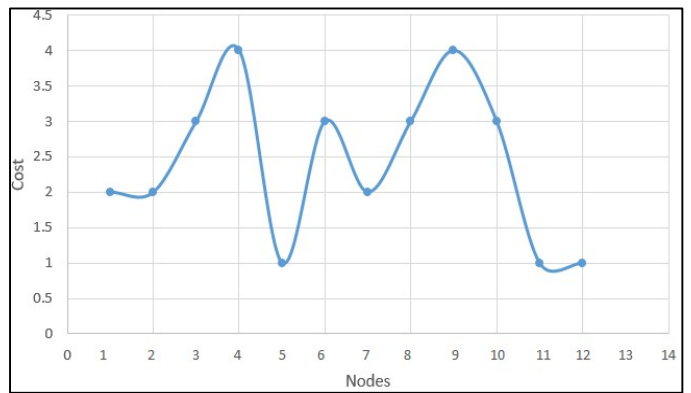


Fig. 8. MST: Cost in minimum spanning tree

a loop, with associated costs. For resulting spanning tree the weights can be analysed to estimate the cost. As compared to diversity of shortest paths from each node to source node, as described in Dijkstra Algorithm, the single shortest path containing all nodes in form of minimum spanning tree is the point of concern. Therefore, this shortest path needs to be taken into account in order to understand the morphology of streets to be locked down.

VI. CONCLUSION

The public health domain deals with numerous infectious diseases in terms of providence of medical assistance, resource providing and defining strategies to control the epidemics. The current outburst of COVID-19 falls under one of categories of infectious diseases and needs to be addressed properly in perspective of providing better resilience plans. Therefore, lockdown has been adopted as a very first step to control the severe outcome of mortalities. To take timely decision for lockdown, concepts of spatial networks along with shortest path algorithms have been used to provide better insight into affected areas. MST has been proved to capture keys areas that can be considered as pivots for spread of infectious disease, hence easing the decision by targeting the specific areas of concern for lockdown.

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